

Homework — Week 10
Partial Derivatives, Level Curves & Tangent Planes

Exercise 1. Find f_x and f_y for

$$f(x, y) = 4x^3y^2 - 7xy^4 + 2x^2 - 5y + 3.$$

Exercise 2. Find f_x and f_y for

$$f(x, y) = e^{3x^2-y} \cdot \ln(xy).$$

Exercise 3. Find f_x and f_y for

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$$

Exercise 4. Find f_x and f_y for

$$f(x, y) = (2x^2y + y^3)^5.$$

Exercise 5. Find f_x and f_y for

$$f(x, y) = x^3 \sin(x^2y).$$

Exercise 6. Find f_x , f_y , and f_z for

$$f(x, y, z) = x^2e^{yz} - z \ln(xy) + y^3z^2,$$

and evaluate each partial derivative at the point $(1, 1, 0)$.

Exercise 7. Find f_x and f_y for

$$f(x, y) = \sqrt{2x^2 + xy + y^2}$$

and evaluate $f_x(1, 2)$ and $f_y(1, 2)$.

Exercise 8. Find and describe the level curves of

$$f(x, y) = \frac{y}{x^2}$$

for $C = -2, -1, 1, 2$.

Exercise 9. Find and describe the level curves of

$$f(x, y) = x^2 - y^2$$

for $C = -4, -1, 0, 1, 4$.

Exercise 10. Find the equation of the tangent plane to each of the following surfaces at the given point.

a) $f(x, y) = x^2y + \cos(xy)$ at $(1, 0, f(1, 0))$.

b) $f(x, y) = \ln(x^2 + 2y^2 + 1)$ at $(0, 0, f(0, 0))$.